

## Conv-ChArT — pose accuracy, four systems

Rotation in **degrees** (deg) — the geodesic angle between true and estimated board rotation. Translation in **board squares** (sq): the evaluation set fixes square\_length\_m = 1.0, so the unit is board squares, not metres. 1500 images.

**All four algorithms share the identical downstream pipeline** — lattice gate, recovery pass, then PnP (IPPE) — so the only thing that differs between rows is the corners each detector produced. Classical OpenCV and Deep ChArUco are therefore *not* using their own pose solvers; these numbers will not match their native output, by design. A **refusal** (a pose the pipeline declines to corroborate) is counted in the solve rate and excluded from the error statistics — it is a correct outcome, not an error.

| algorithm                         | solve %      | rot          |                |               | trans       |              | trans p95 (sq) |
|-----------------------------------|--------------|--------------|----------------|---------------|-------------|--------------|----------------|
|                                   |              | median (deg) | rot mean (deg) | rot p95 (deg) | median (sq) | mean (sq)    |                |
| <b>Conv-ChArT + refiner</b>       | 94.3         | 0.137        | 0.478          | 1.035         | 0.0104      | 0.0415       | 0.1545         |
| <b>Conv-ChArT-FAST + refiner</b>  | 84.7         | 0.130        | 0.958          | 1.050         | 0.0096      | 0.1419       | 0.1193         |
| Deep ChArUco (fine-tuned)         | 94.5         | 0.873        | 10.771         | 90.468        | 0.0775      | 1.2615       | 7.3668         |
| classical OpenCV                  | 54.9         | 0.148        | 0.991          | 2.209         | 0.0191      | 0.0720       | 0.2230         |
| <b>lead: ours vs Deep ChArUco</b> | <b>1.00x</b> | <b>6.4x</b>  | <b>22.5x</b>   | <b>87.4x</b>  | <b>7.4x</b> | <b>30.4x</b> | <b>47.7x</b>   |

### Reading the table

- **The lead row reads as “how many times better our refined arm is”**, computed per column in that column’s own direction: solve % is higher-is-better, every error column is lower-is-better. A value below 1.0 means we are behind on that column. It is quoted against Deep ChArUco only — see the classical note below for why a multiplier against classical would compare two different populations.
- **Every row here is refined except the baselines.** Our coarse (pre-refiner) arm is measured but not shown; the refiner is worth 2.6x on rotation median and 2.6x on translation median, so quoting the refined arm is quoting the system a docking controller actually consumes. **Deep ChArUco’s own RefineNet is not run**, so its row is unrefined — an asymmetry that favours us, stated here rather than buried.
- **Classical OpenCV is accurate when it succeeds** — median rotation 0.148 deg against our 0.137 deg — but it solves only 55% of frames against our 94%. The claim against it is coverage, not precision, and that is exactly why it gets no multiplier row: its error columns are conditioned on the 55% of frames it chose to attempt, which is an easier set than the one we are scored on.
- **Deep ChArUco has a heavy tail.** Its median is respectable, but p95 rotation is 90 degrees: confidently wrong poses on several percent of frames. For a docking robot a wrong pose is more dangerous than no pose, which makes the p95 column more operationally important than the median.
- **Solve rate and p95 must be read together.** Solve rate alone flatters Deep ChArUco; p95 alone hides that classical refuses nearly half the time.